**Ex. No.: 6**
**IPC USING SHARED MEMORY**

**Aim:**
To write a C program to perform Inter Process Communication (IPC) using shared memory between a sender process
and a receiver process.

**Algorithm:**
**Sender:**
1. Set the size of the shared memory segment.
2. Allocate the shared memory segment using shmget().
3. Attach the shared memory segment using shmat().
4. Write a string to the shared memory using sprintf().
5. Introduce delay using sleep().
6. Detach the shared memory using shmdt().

**Receiver:**
1. Set the size of the shared memory segment.
2. Allocate the shared memory segment using shmget().
3. Attach the shared memory segment using shmat().
4. Read and print the message from shared memory.
5. Detach the shared memory using shmdt().

**Program Code:**
sender.c
```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
int main() {
  int shmid;
  char *str;
  shmid = shmget(1234, 1024, 0666 | IPC_CREAT);
  str = (char*) shmat(shmid, NULL, 0);
  sprintf(str, "Hello from Sender Process!");
  printf("Data written to shared memory: %s\n", str);
  sleep(5);
  shmdt(str);
  return 0;
}
```

receiver.c
```
#include <stdio.h>
#include <sys/ipc.h>
```

```
#include <sys/shm.h>
int main() {
  int shmid;
  char *str;
  shmid = shmget(1234, 1024, 0666);
  str = (char*) shmat(shmid, NULL, 0);
  printf("Data read from shared memory: %s\n", str);
  shmdt(str);
  return 0;
}
```

**Output:**

Sender Output: Data written to shared memory: Hello from Sender Process!
Receiver Output: Data read from shared memory: Hello from Sender Process!

**Result:**

Thus the program to perform interprocess communication using IPC has been successfully executed.